

The Death of Mystery Has Been Greatly Exaggerated: Comments on Sripada et al. (2026)

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Abstract

In response to Futrell & Mahowald's (2025) paper on the role of Large Language Models (LLMs) in linguistics, Sripada et al. (2026) strikingly argue that LLMs "illuminate" the mechanisms underlying what Chomsky calls the creative aspect of language use (CALU). Such illumination promises, they argue, to bring this once-intractable problem under the heel of scientific inquiry by advancing our understanding of the mechanisms by which individuals select situation-appropriate expressions from an unbounded class of expressions. Sripada et al. draw attention to an important, foundational issue in generative linguistics, but their paper mischaracterizes CALU. The mischaracterization turns principally on the notion of stimulus-independence and how this relates to the capacity for unbounded linguistic production. These comments seek to correct this mischaracterization by showing that the problem of CALU is deeper and more distinctive than they let on. Likewise, misunderstandings of Chomsky's invocation of CALU are clarified.

Introduction

In response to a *Behavioral and Brain Science (BBS)* article by Futrell & Mahowald (2025) on the role of Large Language Models (LLMs) in linguistics, Sripada et al. (2026)¹ make a striking claim: beyond the use of LLMs for better understanding human syntax, LLMs can be used to “illuminate” the “deepest enigma about language: what Chomsky called the creative aspect of language use (CALU)” (65).

Sripada et al. characterize CALU as a “speaker’s ability to freely select, from the vast array of grammatically possible sentences, utterances that are appropriate to the situation and linguistic context” (Sripada et al., 65). Here, they reference the “stimulus-independence” of language use traceable back to Descartes, in this way “[transcending] stimulus-response associations” (65).

The authors likewise note that Chomsky adopted a mysterian approach to this matter, for which “LLMs pose a stark challenge.” The challenge, simply put, is that LLMs may allow scholars to at last inch toward mechanistic understanding of CALU due to these models’ ability to “generate fluent, contextually appropriate responses” along with “humanlike levels of novelty” (65). They posit four LLM mechanisms that are plausibly relevant to this end: (1) A vast features space that encodes meaning and contextual boundaries, retrievable and combinable for appropriate response; (2) Attention mechanisms for context-sensitivity; (3) Incremental prediction based on autoregressive generation for novelty; and (4) Reinforcement Learning for situational goals and pragmatic appropriateness (65).

Equally striking, the authors attribute the alleged success of LLMs in this respect to “an emergent property of *scale*: it arises only when an architecture has access to extraordinarily rich representational sources and correspondingly powerful mechanisms for manipulating them” (65).

The authors’ invocation of CALU is laudable - LLMs, at least at first glance, motivate raising the question of whether these systems replicate the ability to *use language creatively*. They also rightly recognize that the issue of appropriate language use, in this sense, was present in Chomsky’s work “as early as his 1959 review of Skinner’s *Verbal Behavior*.” It is here, they argue, that “LLMs shift the boundary of tractability” (66). Likewise, the Sripada et al.’s identification of the four LLM mechanisms may indirectly aid scholars’ characterization of ordinary human language use, albeit not its explanation (see below).

Functional vs. Human Appropriateness

That said, the authors diminish the *distinctiveness* of the problem of creative language use, as Chomsky characterizes it, while also exaggerating the *significance* of LLMs in this respect.

Both Chomsky and fellow travelers have made clear over time that CALU is a “special” problem only in light of ordinary language use simultaneously meeting three conditions: (1) Stimulus-freedom, or the ability to use language independently of identifiable stimuli in the local environment; (2) Unboundedness, or the ability to produce and understand an infinite array of structured expressions; and (3) Appropriateness to situations, as judged by other human participants in the discourse situation. For behavior to be “creative,” in this sense, all three characteristics must be present (see, Baker, 2007, 236-237; Collins, 2021, 560).

¹ All references to Sripada et al. are to Sripada et al. (2026).

Although Sripada et al. explicitly invoke the “stimulus-independence” of CALU (65), the remainder of their brief commentary proceeds to neglect to ask whether LLMs are stimulus-free. This does not appear to be the case. As Carchidi (2024; 2026b) argue, LLMs output content (whether that content is natural language or not) in fixed relation with external stimuli, identifiable independent of the particulars of the situation (whether we deem such behavior “intelligent” and so forth is irrelevant). Chomsky, in his critique of Skinner, indeed takes issue with the notion of stimulus-control on the following grounds:

We identify the stimulus when we hear the response...the talk of stimulus control simply disguises a complete retreat to mentalistic psychology. We cannot predict verbal behavior in terms of the stimuli in the speaker’s environment, since we do not know what the current stimuli are until he responds (Chomsky, 1959, 52).

For Chomsky, stimulus control as an explanatory concept was stretched by Skinner in its application such that it was voided of content: “the word stimulus has lost all objectivity in this usage. Stimuli are no longer part of the outside physical world; they are driven back into the organism” (Chomsky, 1959, 52).

This problem does not arise in the case of LLMs, most obviously in the form of chatbots. Outputs are fixed to stimuli in ways that do not require post-hoc attribution, or some other “interpretation of an event as part of a pattern” of the discourse (or other) situation (McGilvray, 2001, 7). It is more accurate to suggest that LLMs exhibit a *functional appropriateness* – the alignment of outputs with the expectations and intents of a human end-user, successively improving generation-after-generation by shaping the distribution from which LLMs sample in conformance with their uses as extensions of ourselves.² Because these outputs are stimulus-fixed, their appropriateness can be distinguished from “the much more obscure relation of appropriateness” that governs human language use, making human language “available for the free expression of thought precisely because it is not tied directly to external stimuli” (Chomsky, 1975, 302).

It is also interesting to note that fellow travelers in generative linguistics have taken pains to make the distinction between functional and human appropriateness. McGilvray indeed noted that the stimulus-freedom of language use entails that being “appropriate” cannot be tantamount to “being caused by environmental conditions” (McGilvray, 2001, 9). Such behavior is, as in the case of LLMs, functional in that it is goal-oriented, or otherwise tethered to external stimuli (McGilvray, 2001, 8-10).

It is conceivable that Sripada et al. mean something somewhat different by CALU than Chomsky and other generative linguists due to a different interpretation of Chomsky’s critique of Skinner. In particular, whereas the explication of stimulus-freedom in subsequent years was bound up in a claim about the inability to identify a set of stimuli on which one can build a science of language use, Sripada et al. appear to mean the narrower break from stimulus-response psychology in which utterances are not mechanically matched with stimuli; the *novelty* of appropriate language use in relation to stimuli being the point. LLMs, capable of appropriate novelty, appear to therefore replicate CALU, for Sripada et al.

In this way, Sripada et al. (and Futrell & Mahowald (2026) in extension) appear to make a mistake adjacent to a longstanding confusion about Chomsky’s invocation of the term “creative aspect of language use.” Chomsky has indeed lamented: “[A] number of professional linguists have repeatedly confused what I refer to here as “the creative aspect of language use” with the recursive property of generative

² On the use of machines as extensions of human beings, see, McDonough (1994, 126).

grammars, a very different matter” (Chomsky, 2006, xiv). For the reasons stated, this is a narrower notion of what is meant by CALU, though I will not speculate further on this point. Suffice it to say, for Chomsky, the term *stimulus-free* and the term *uncaused* are used equivalently (McGilvray, 2005, 220-221).

Mystery Made Tractable?

In their response article, Futrell & Mahowald hint at multimodal LMs that “are ready-made for grounding,” sharing Sripada et al.’s optimism for making CALU tractable. They argue: “In that sense, “saying the right thing” can become a matter of saying the right thing conditioned on a particular non-linguistic state” (Futrell & Mahowald, 2026, 78). This, however, shifts the target of inquiry: the distinction between “saying the right thing” in arbitrary, non-experimental contexts and “saying the right thing conditioned on a particular non-linguistic state” is the distinction between the study of *creative* linguistic behavior (in Chomsky’s sense) and *functional* linguistic behavior. Sripada et al., to be sure, correctly noted that the intractability was one motivation for the competence-performance distinction, yet it is unclear that their proposal – with all three characteristics of CALU in mind in relation to LLMs – does not reduce to a proposal to study the mechanisms that *enable* creative linguistic behavior in humans (competencies), rather than the behavior itself (performance), thereby returning us to Chomsky’s position.

This slippage occurs in the notion of the ‘selection’ of an utterance, too. The problem that CALU poses is how an individual *selects* from an unbounded class of utterances in ways that are *appropriate* to a situation but not *caused* by the situation. By narrowing one’s research agenda to the mechanisms that *enable* an individual to produce expressions that fit a situation that causes them, Sripada et al. appear to default back into the distinction between mechanisms (competence) and behavior (performance) that has motivated the characterization of this problem as intractable.

To be sure, the four LLM mechanisms and concomitant emphasis on scale that Sripada et al. highlight may be useful in clarifying our mechanistic understanding of CALU’s *bases* within the human mind (albeit, not the behavior itself). It may be true that only with, for instance, something comparable to attention mechanisms in transformers that creative language use can be undertaken, a compatibility with Merge and Minimalism noted by Marcolli et al. (2025, 336).³ If one squints, Chomsky can also sometimes be understood as suggesting that scale – of the right kind – introduces the characteristics of the human mind he deems potentially mysterious: “Descartes also arrived, quite early in his investigations, at the conclusion that the study of mind faces us with a problem of quality of complexity, not merely degree of complexity” (Chomsky, 2006, 5).

Although neither Sripada et al. nor Futrell & Mahowald discuss agentic LLMs (in this specific exchange), I will gently suggest, on the basis of in-progress work (Carchidi (2026a)), that agentic LLMs *extend* the underlying *productive capacities* of LLMs – “the power to combine a given set of elements in accord with a given system or set of rules in order to get a ‘new’ structure” (McDonough, 1993, 39) - but neither

³ While taking pains, to be sure, to note that this compatibility does not necessarily mean LLMs and human brains learn how to link syntax and semantics in the same ways (Marcolli et al., 2025, 336-337).

replicate the creative aspect of language use defined as the *stimulus-free and unbounded expression of appropriate thought*.⁴

Freedom and Language Use

Even among some generative linguists, Chomsky's position on CALU has often been conflated with other matters related to the intractability of complicating performance factors in the course of ordinary human behavior that justify a competence-performance distinction.⁵ However, what distinguishes Chomsky's near-mysterian position is that *creativity* is a technical notion that characterizes the human mind as a "mind with free will" (McGilvray, 2005, 222). In this vein, Chomsky has at times questioned the specification of the notion of *cause* as applied to behavior, even suggesting that "we have little reason to believe that normal behavior *is* caused, at least in any known sense of the term, nor would a methodological naturalist dogmatically assume otherwise" (Chomsky, 1994, 199). He has elsewhere appeared to reject the notion that *reasons* must be construed as *causes* (see, Chomsky, 2014, 138-139).

Likewise, some pushback on the framing of CALU in the *BBS* exchange is necessary. Futrell & Mahowald (2026, 78) indicate that CALU is "an old and somewhat dormant idea in linguistics," while Sripada et al. (65) indicate that Chomsky first "envisioned an intimate connection between generative linguistics and CALU...But he later came to endorse a mysterian stance regarding CALU as a whole."

These statements are mostly inaccurate. Although Chomsky's (1965, 3-6) earliest invocation of CALU appeared to lump this behavior into linguistic competence, this sloppiness (or simply incomplete formulation) was quickly remedied in his *Cartesian Linguistics*, where Chomsky (1966, ch. 1) laments the inattention to the observations made by the Cartesians in the seventeenth century that constitute CALU. Chomsky has so frequently introduced this idea in his subsequent introductions to general matters in linguistics, one wonders how it could be missed. Chomsky indeed invokes CALU as part of the "Galilean Challenge" (Chomsky, 2017), from which the segmentation of competence from performance is derived. Finally, it is unclear that Chomsky *is* a mysterian with respect to CALU. Although interviews with Chomsky sometimes give this impression, his extended statements on the philosophy of mind and cognitive science have tended to be more cautious (see, e.g., Chomsky, 2009; see also, Collins, 2021).

All this to say: the problem that CALU has traditionally posed in generative linguistics is an old and thorny problem, given a particular gloss by Chomsky, and plays a rather specific limiting role in that approach's conception of a science of language. It is unclear that LLMs either replicate this behavior or shed light on how we might advance our understanding of this behavior in humans. It *may* aid our understanding of the mechanisms of selection that underpin stimulus-free and novel yet appropriate behavior, although this should crucially be understood as retaining the intractability of the behavior itself that has motivated generative linguists to, among other things, invoke a competence-performance distinction. Scholars should be exceptionally careful about how one characterizes the relation between an LLM's outputs and its environment, and a human's use of language and their environment.

⁴ What has repeatedly been referred to as the "creative" abilities of LLMs, both in common parlance and seemingly in Sripada et al.'s paper, is in fact productivity. Part of the trick is in clarifying the relation of an LLM's productive capacities to its environment, and this relation appears firmly functional over time and across generations.

⁵ This is not to say this is not one such justification. For an explication, see, Volenec & Reiss (2025, 4-5). On the relation of this reasoning to the "Galilean" style of explanation, see, Rey (2020, ch. 1).

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